

Title

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OUTLINE



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EXECUTIVE SUMMARY



- **Project Objective**
 - Analyze developer demographics and technology trends.
 - Identify insights from the Stack Overflow Developer Survey.
 - Present results through interactive dashboards.
- **Data Analysis Overview**
 - Explored demographics by age, gender, education, and country.
 - Analyzed current technology usage.
 - Compared current and future technology preferences.
- **Dashboard Development**
 - Created dashboards using multiple chart types.
 - Applied filters and visual features to improve clarity.
- **Key Findings**
 - Identified the most preferred technologies.
 - Revealed demographic and geographic patterns.
 - Highlighted future technology trends.
- **Conclusion**
 - The analysis provides valuable insights into developer preferences.
 - Results support informed learning, career, and technology decisions.



INTRODUCTION



- **Purpose of the Report**
 - Analyze the Stack Overflow Developer Survey dataset.
 - Identify trends in developer demographics and technology preferences.
- **Target Audience**
 - Students and aspiring developers.
 - Technology professionals and employers.
 - Organizations interested in technology trends.
- **Value of the Analysis**
 - Supports data-driven decision-making.
 - Identifies emerging technologies and developer preferences.
 - Provides insights for career planning and technology adoption.

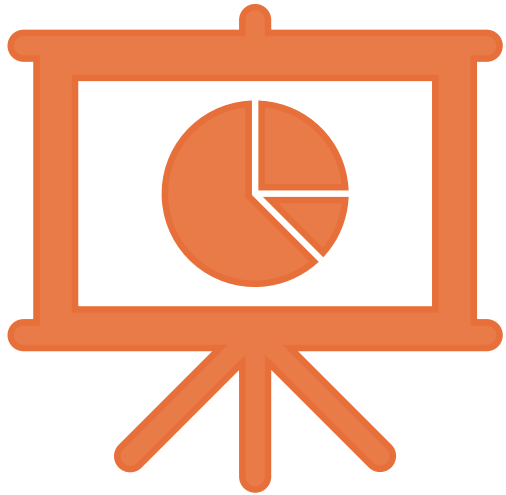
METHODOLOGY



- **Data Sources**
 - Used the Stack Overflow Developer Survey dataset.
 - Dataset contains information on developer demographics, skills, and technology preferences.
- **Data Collection**
 - Collected survey responses from developers worldwide.
 - Imported the dataset into the analytics tools for analysis.
- **Data Wrangling Steps**
 - Cleaned and prepared data for visualization.
 - Removed null values and standardized categories.
 - Organized fields for dashboard creation.



Results



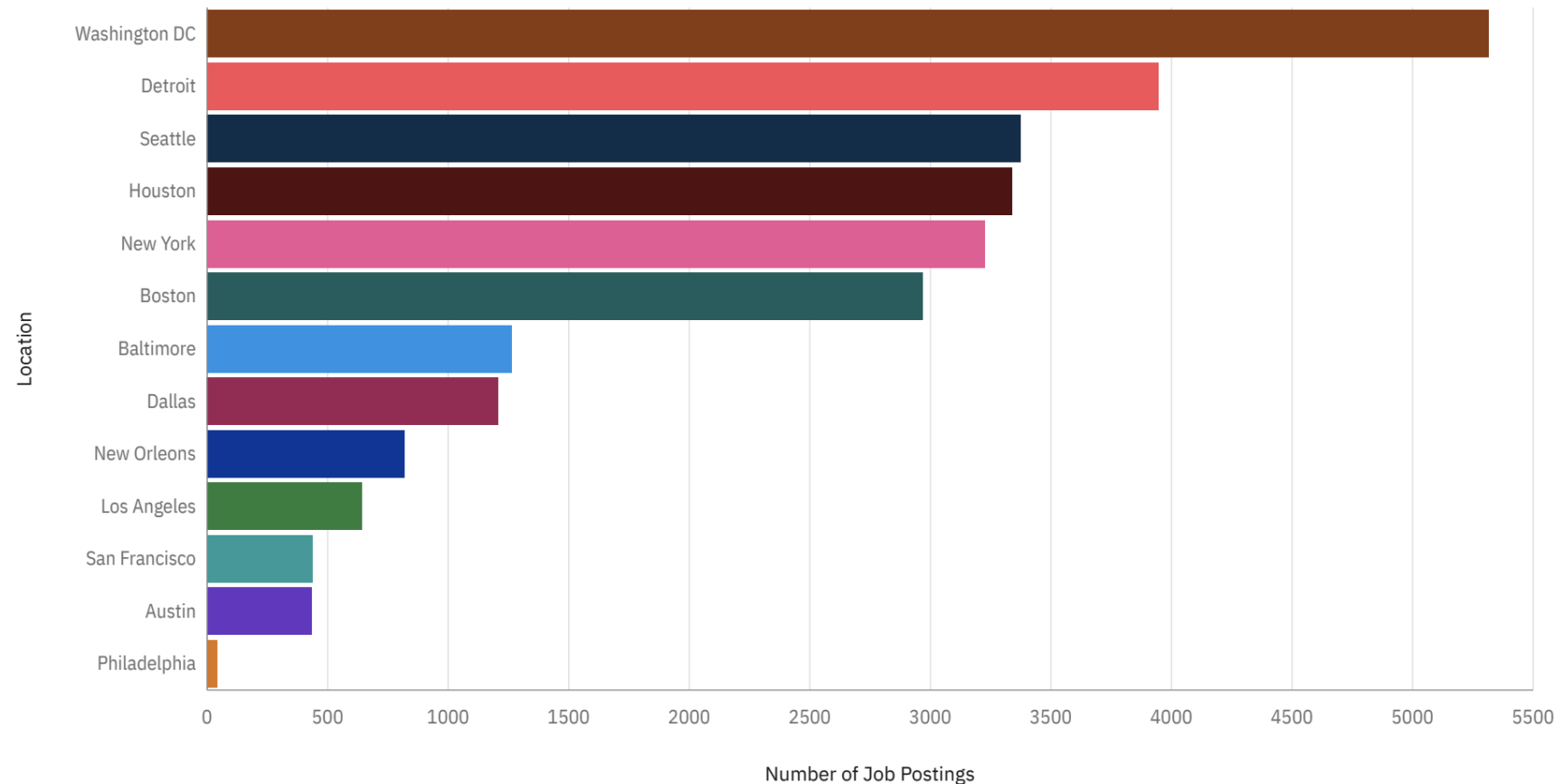
<Please present your results in the following slides.>



JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.

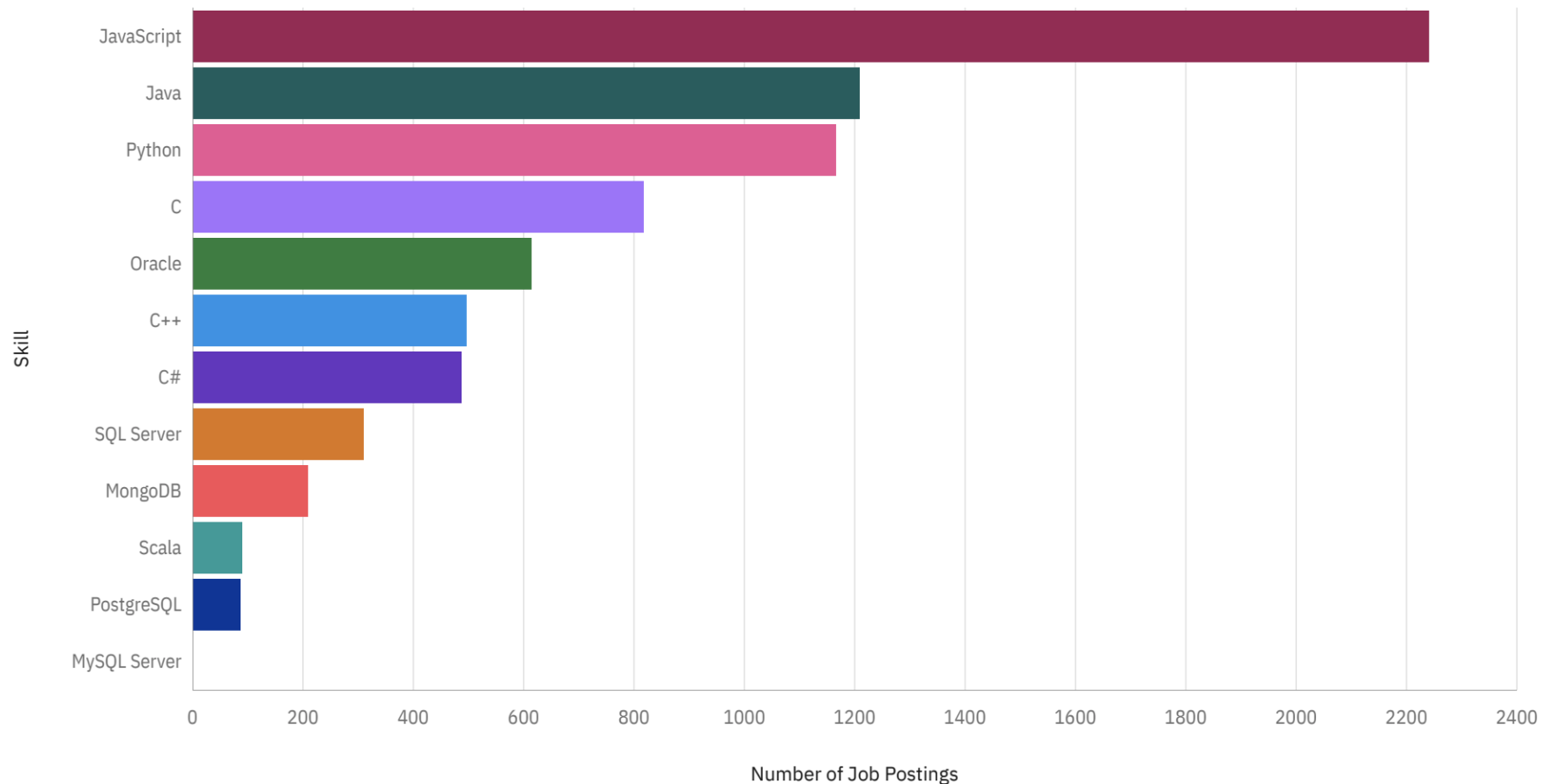
Number of Job Postings by Location



POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.

Number of Job Postings by Skill

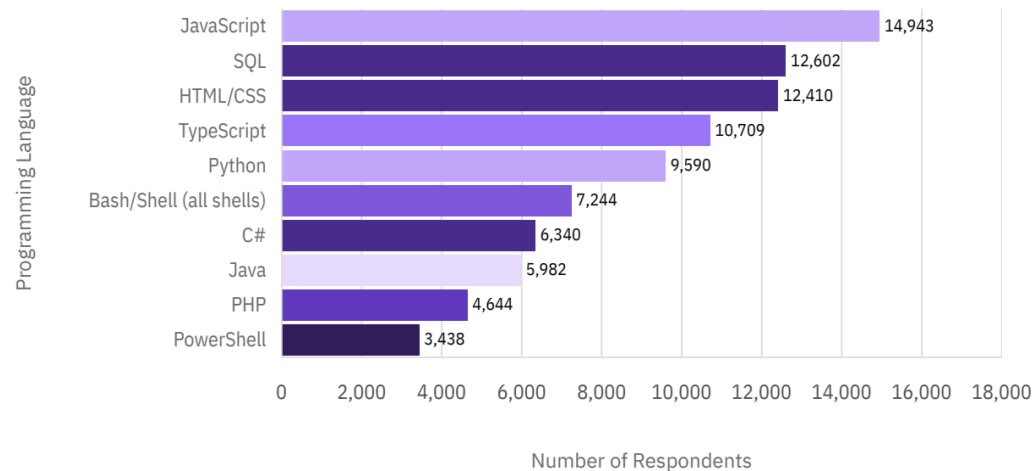


PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year

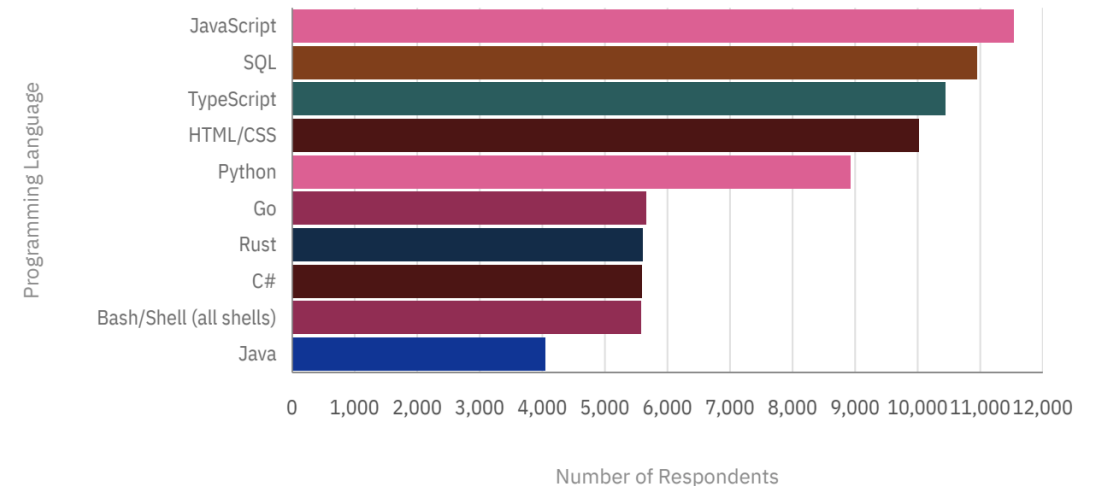
Top 10 Programming Languages Used by Respondents



< Bar chart of top 10 programming languages for the current year.>

Next Year

Top 10 LanguageWantToWorkWith.



< Bar chart of top 10 programming languages for the next year.>



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **JavaScript shows the strongest overall adoption trend**, ranking first in both current usage (**14,943 respondents**) and developer interest (**11,541 respondents**), indicating continued importance in software development.
- **Established technologies such as SQL, HTML/CSS, TypeScript, and Python remain highly relevant**, appearing among the top languages developers currently use and want to continue working with.
- **Emerging languages such as Go and Rust show growth potential**, ranking highly in developer preferences despite having lower current usage compared with more established languages.

Implications

- **Organizations should continue investing in JavaScript-based technologies**, as strong current usage and continued interest indicate long-term relevance.
- **Companies should maintain demand for skills in Python, SQL, TypeScript, and web technologies**, as these languages support important areas such as data, backend, and application development.
- **Technology teams should monitor adoption of Go and Rust**, as increasing developer interest may create future opportunities in areas requiring scalability, performance, and security.

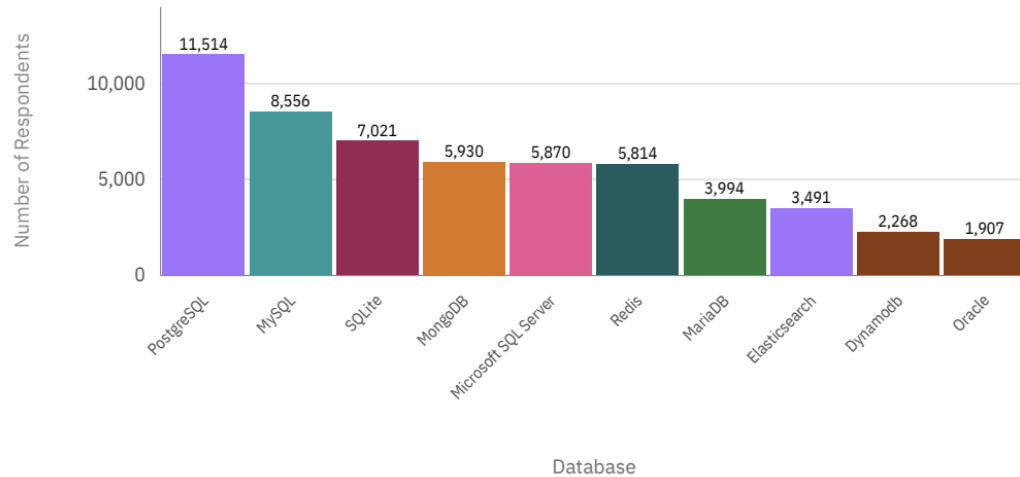


DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year

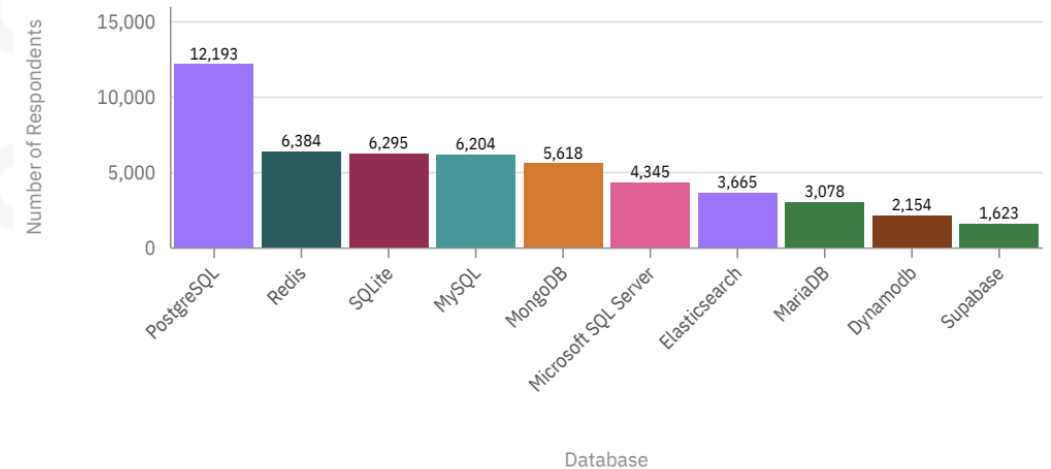
Top 10 Databases Used by Respondents



< Bar chart of top 10 databases for the current year goes here >

Next Year

Top 10 Most Desired Databases to Work With



< Bar chart of top 10 databases for the next year goes here.>



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **PostgreSQL shows the strongest overall adoption trend**, ranking first in both current usage (**11,514 respondents**) and developer interest (**12,193 respondents**), indicating strong acceptance and continued demand.
- **Relational databases remain highly relevant**, with technologies such as MySQL, SQLite, and Microsoft SQL Server appearing among the most used databases and maintaining strong developer interest.
- **Developers show growing interest in modern and specialized databases**, with Redis, MongoDB, and Elasticsearch ranking highly in desired technologies, reflecting demand for flexible and scalable data solutions.

Implications

- **Organizations should prioritize PostgreSQL expertise**, as its strong current adoption and developer interest suggest it will remain an important database technology.
- **Companies should continue supporting relational database skills**, as SQL-based systems remain widely used for business applications and data management.
- **Technology teams should evaluate modern database solutions such as Redis, MongoDB, and Elasticsearch**, as their popularity indicates increasing demand for performance, scalability, and specialized data processing capabilities.



DASHBOARD



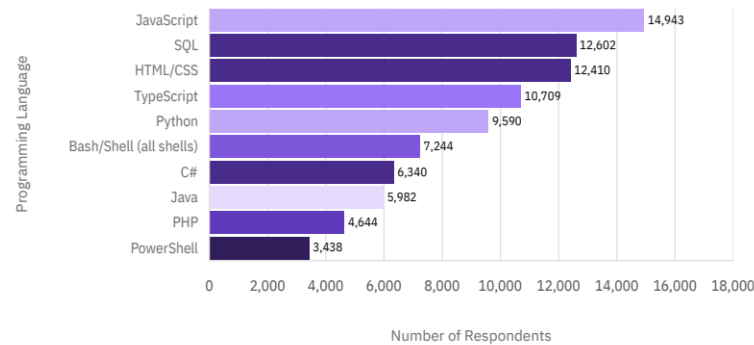
<Please present your dashboard in the following slides.>

DASHBOARD TAB 1: Current Technology Usage

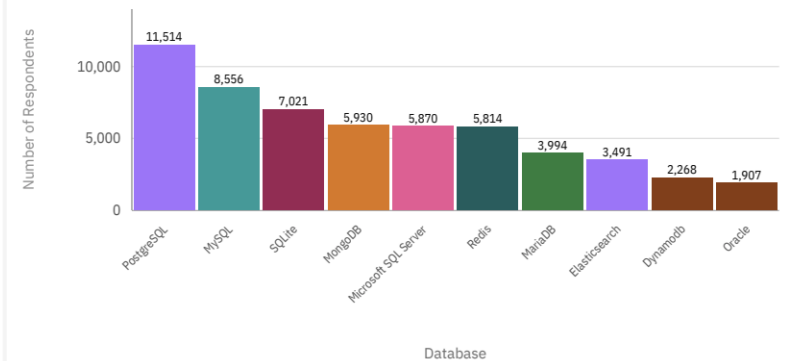
- **JavaScript and PostgreSQL** are the most widely used programming language and database, highlighting their strong adoption among developers.
- **Amazon Web Services (AWS)** is the most commonly used cloud platform, indicating its leadership in cloud computing.
- **React and Node.js** are the most widely used web frameworks, demonstrating their continued popularity in modern web development.

Current Technology Usage

Top 10 Programming Languages Used by Respondents



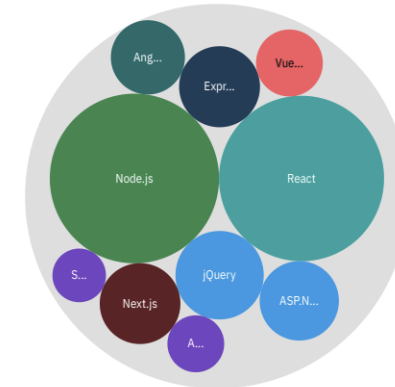
Top 10 Databases Used by Respondents



Top 10 Platforms Used by Respondents



Top 10 Web Frameworks Used by Respondents

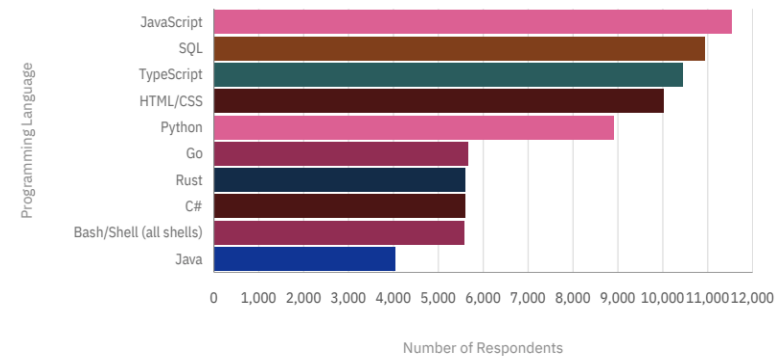


DASHBOARD TAB 2: Future Technology Trends

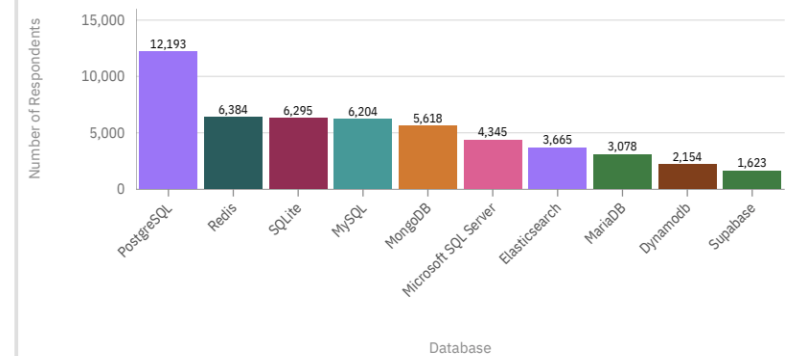
- **JavaScript** remains the most desired programming language, while Go and Rust appear among the top technologies developers want to work with.
- **PostgreSQL** continues to lead developer interest, with Redis and SQLite also showing strong demand among database technologies.
- **React** and **Node.js** remain highly desired frameworks, indicating sustained interest in modern JavaScript-based development ecosystems.

Future Technology Trend

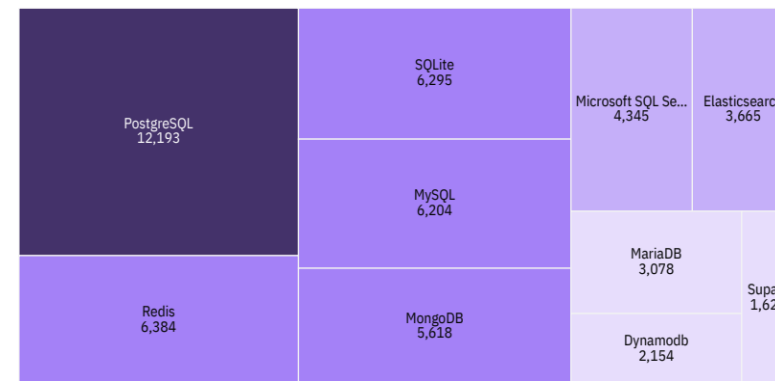
Top 10 LanguageWantToWorkWith.



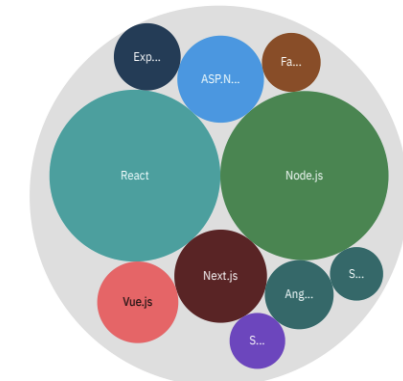
Top 10 Most Desired Databases to Work With



Top 10 Most Desired Platforms to Work With



Top 10 Most Desired Web Frameworks to Work With



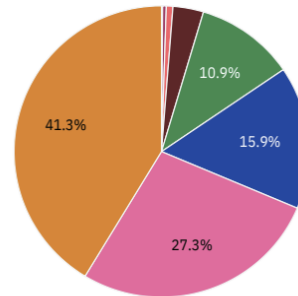
DASHBOARD TAB 3: Demographics

- **Most respondents are between 25–34 years old, followed by the 35–44 age group**, indicating a predominantly early- to mid-career developer population.
- **Bachelor's degrees are the most common education level**, with master's degrees representing the second-largest group of respondents.
- **Survey responses are globally distributed**, with the highest concentration of respondents coming from the United States, reflecting broad international participation.

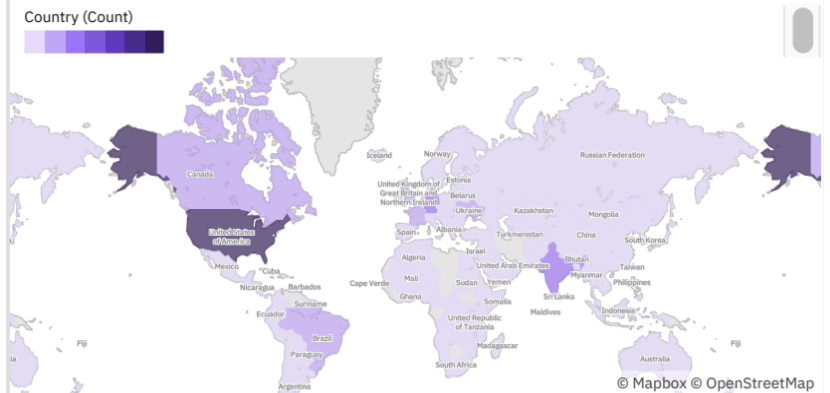
Demographics

Respondent Distribution by Age

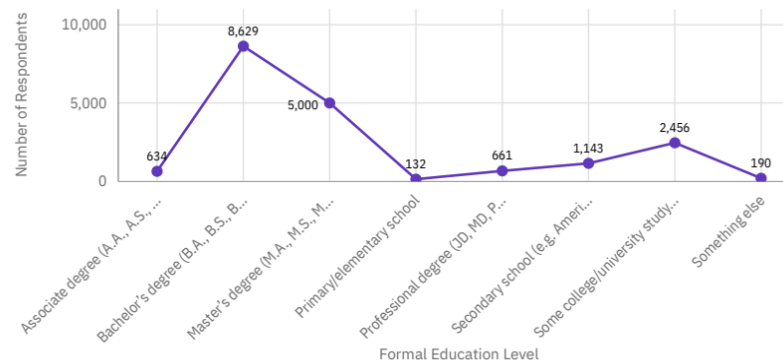
● Prefer not to say ● 65 years or older ● Under 18 years old ● 55-64 years old ● 45-54 years old ● 18-24 years old
● 35-44 years old ● 25-34 years old



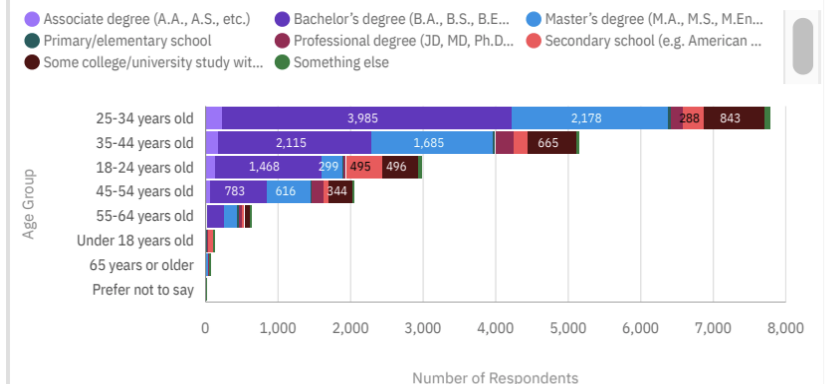
Respondent Count by Country



Respondent Distribution by Formal Education Level



Respondent Count by Age and Education Level



DISCUSSION

- Discuss overall insights derived from dashboards



OVERALL FINDINGS & IMPLICATIONS

Findings

- **JavaScript, PostgreSQL, React, and AWS are the leading technologies**, demonstrating strong adoption across programming languages, databases, web frameworks, and cloud platforms.
- **Developers continue to favor modern and scalable technologies**, with PostgreSQL, Go, Rust, and React showing strong interest for future adoption.
- **The survey population is primarily composed of developers aged 25–34 with bachelor's degrees**, providing insights into the preferences of an early- to mid-career workforce.

Implications

- **Organizations should continue investing in widely adopted technologies** such as JavaScript, PostgreSQL, React, and AWS to meet current industry demands.
- **Companies should prepare for evolving technology trends** by building expertise in emerging technologies like Go, Rust, and modern cloud-native solutions.
- **Training and workforce development programs should align with current developer preferences**, emphasizing in-demand programming, database, cloud, and web development skills.

CONCLUSION



- The survey highlights **JavaScript, PostgreSQL, React, and AWS** as the most widely adopted technologies, reflecting their importance in today's software development landscape.
- Developer preferences indicate **continued demand for established technologies** while showing increasing interest in emerging tools such as Go and Rust.
- **The demographic profile** suggests that technology trends are primarily driven by early- to mid-career professionals with **strong formal education backgrounds**.
- These insights can help organizations prioritize technology adoption, workforce development, and strategic planning to remain competitive in the evolving technology industry.



APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.

